© CFA Institute. For candidate use only. Not for distribution.

Harmonic Mean Formula. The harmonic mean of a set of observations X_1, X_2, …, X_n is:

$$\overline{X}_H = \frac{n}{\sum_{i=1}^{n}(1/X_i)}, \tag{4}$$

with $X_i > 0$ for $i = 1, 2, \ldots, n$.

The harmonic mean is the value obtained by summing the reciprocals of the observations,

$$\sum_{i=1}^{n}(1/X_i),$$

the terms of the form $1/X_i$, and then averaging their sum by dividing it by the number of observations, n, and, then finally, taking the reciprocal of that average,

$$\frac{n}{\sum_{i=1}^{n}(1/X_i)}.$$

The harmonic mean may be viewed as a special type of weighted mean in which an observation's weight is inversely proportional to its magnitude. For example, if there is a sample of observations of 1, 2, 3, 4, 5, 6, and 1,000, the harmonic mean is 2.8560. Compared to the arithmetic mean of 145.8571, we see the influence of the outlier (the 1,000) to be much less than in the case of the arithmetic mean. So, the harmonic mean is quite useful as a measure of central tendency in the presence of outliers.

The harmonic mean is used most often when the data consist of rates and ratios, such as P/Es. Suppose three peer companies have P/Es of 45, 15, and 15. The arithmetic mean is 25, but the harmonic mean, which gives less weight to the P/E of 45, is 19.3.

The harmonic mean is a relatively specialized concept of the mean that is appropriate for averaging ratios ("amount per unit") when the ratios are repeatedly applied to a fixed quantity to yield a variable number of units. The concept is best explained through an illustration. A well-known application arises in the investment strategy known as **cost averaging**, which involves the periodic investment of a fixed amount of money. In this application, the ratios we are averaging are prices per share at different purchase dates, and we are applying those prices to a constant amount of money to yield a variable number of shares. An illustration of the harmonic mean to cost averaging is provided in Example 6.

EXAMPLE 6

Cost Averaging and the Harmonic Mean

1. Suppose an investor invests EUR1,000 each month in a particular stock for $n = 2$ months. The share prices are EUR10 and EUR15 at the two purchase dates. What was the average price paid for the security?

   Solution:

   Purchase in the first month = EUR1,000/EUR10 = 100 shares.

   Purchase in the second month = EUR1,000/EUR15 = 66.67 shares.

   The investor purchased a total of 166.67 shares for EUR2,000, so the average price paid per share is EUR2,000/166.67 = EUR12.
   The average price paid is in fact the harmonic mean of the asset's prices at the purchase dates. Using Equation 5, the harmonic mean price is 2/[(1/10) + (1/15)] = EUR12. The value EUR12 is less than the arithmetic mean purchase price (EUR10 + EUR15)/2 = EUR12.5.